

Customer Shopping Behavior Analysis

Python • SQL • Power BI — end-to-end analytics





Project Overview

Goal

Transform raw transactions into business insights

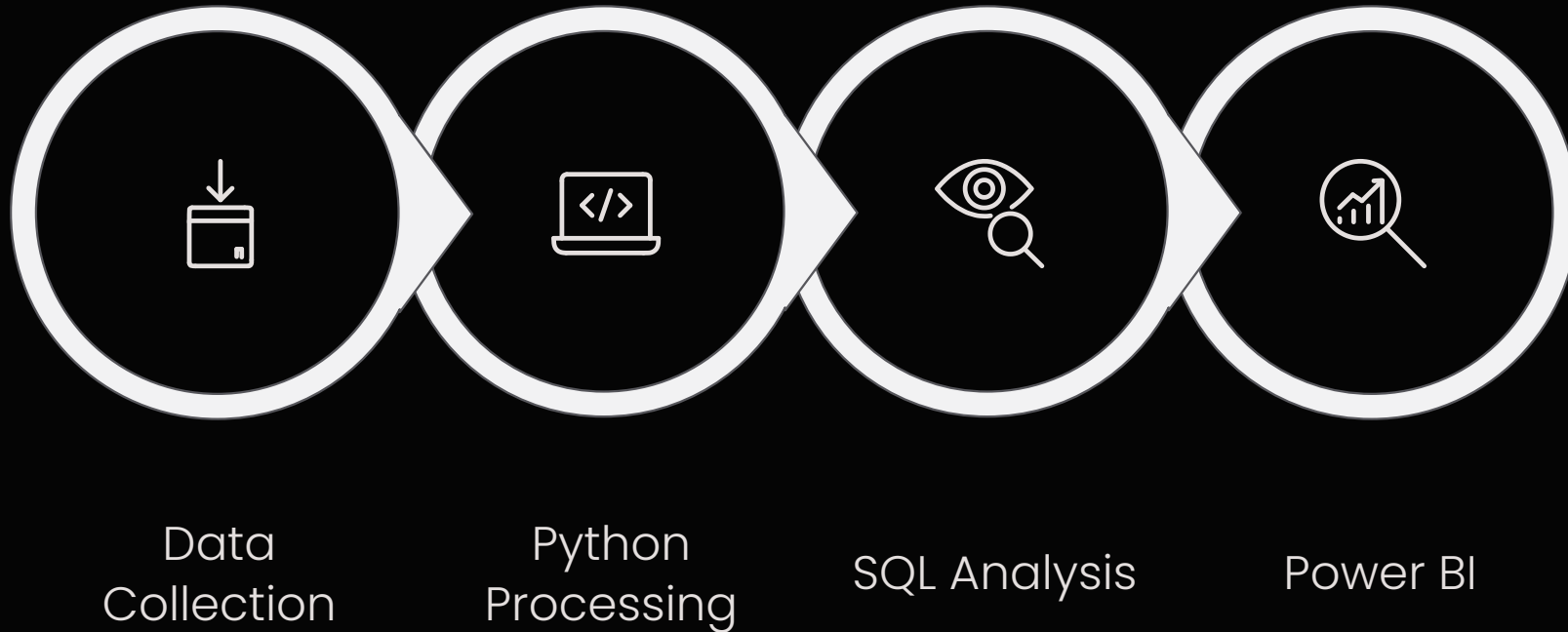
Tech Stack

Python, MySQL, Power BI

Audience

Executives, managers, marketing, analysts

Workflow



End-to-end: collect → clean & feature-engineer → SQL metrics → interactive dashboards

Data Processing (Python)



Cleaning

Missing values, duplicates removed



Feature Engineering

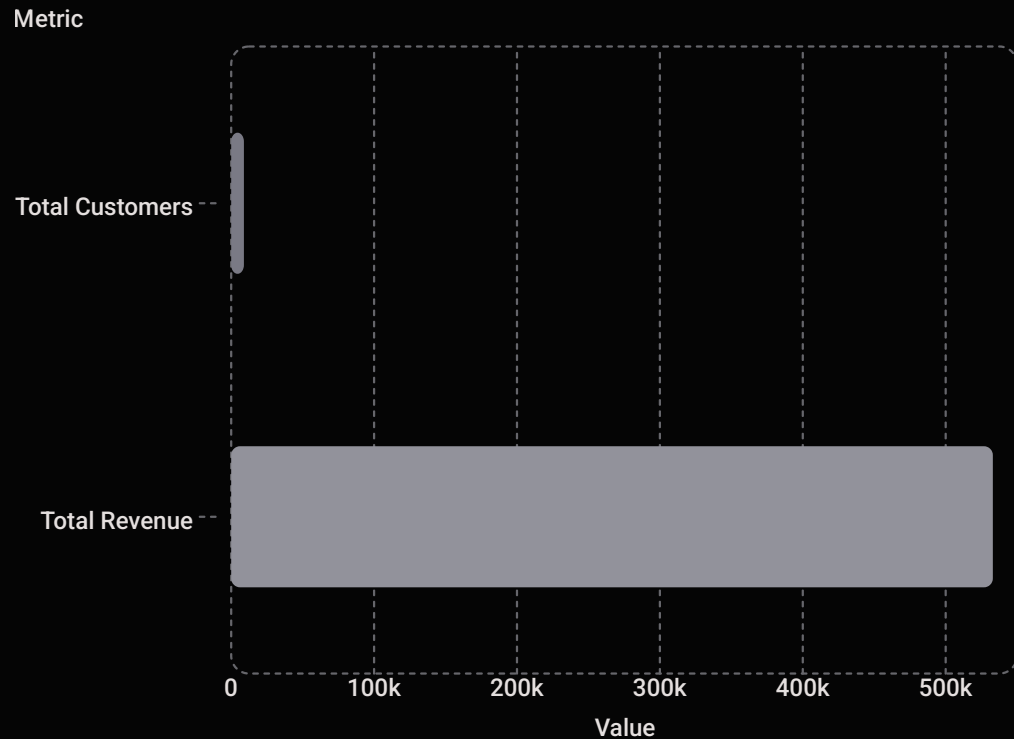
Customer segments, age groups, freq. metrics



Export

Clean dataset exported to MySQL

SQL Analysis — Key Metrics



SQL produced business-ready metrics used in Power BI: revenue breakdowns, loyalty, top products, shipping & discount analysis.



Metrics drive dashboard KPIs and filters

Executive Sales Overview — Highlights



Top Category

Clothing — highest revenue (239K)



2nd Category

Accessories — 169K



Footwear

81K

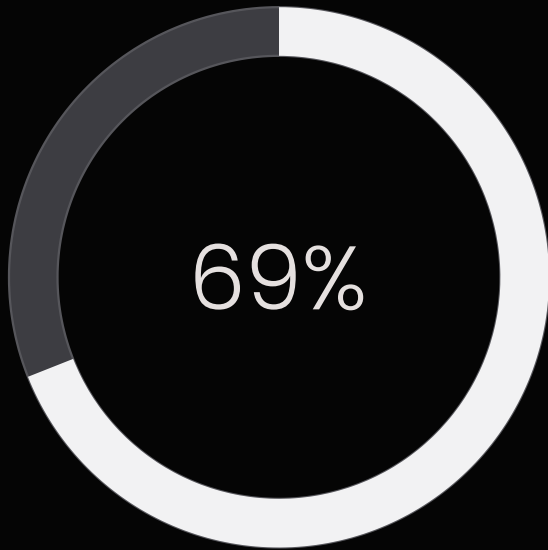


Outerwear

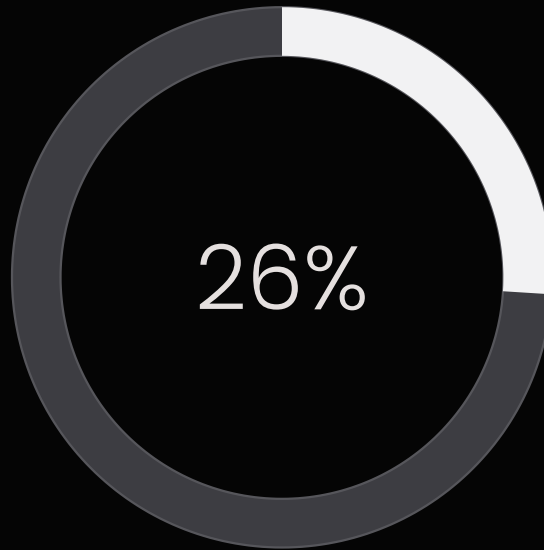
43K

Male customers: 67.61% of revenue • Young Adults top age segment

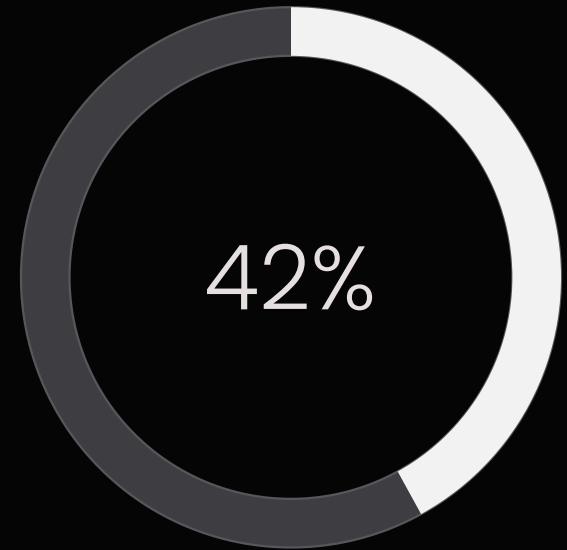
Customer Behavior & Loyalty



Loyal customers



Subscribed



Discount adopters

Common frequencies: monthly, quarterly, annually. Focus on retention — repeat buyers drive revenue.

Product Performance, Findings & Recommendations



- Top product: Blouse
- Highest rated: Gloves
- Standard shipping = best satisfaction
- Total revenue ≈ \$533K from 8.9K customers

Recommendations

Invest in Clothing & Accessories; target Young Adults; grow subscriptions; personalize retention offers; optimize Fall inventory.